

Number Patterns programs in C Language

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Pattern	Logic
<pre>Enter No Of Rows: 5 1 2 3 4 5 1 2 3 4 1 2 3 1 2 1</pre>	<pre>#include <stdio.h> void main() { int rows, i, j; printf("Enter No Of Rows:"); scanf("%d", &rows); for(i = rows; i >= 1; i--) { for(j = 1; j <= i; j++) { printf("%d ", j); } printf("\n"); } }</pre>
<pre>Enter No Rows:5 5 5 5 5 5 4 4 4 4 3 3 3 2 2 1</pre>	<pre>#include <stdio.h> void main() { int rows, i, j; printf("Enter No Rows:"); scanf("%d", &rows); for(i = rows; i >= 1; i--) { for(j = 1; j <= i; j++) { printf("%d ", i); } printf("\n"); } }</pre>



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Pattern

```

    1
  1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

Logic

```
#include <stdio.h>
#include <stdlib.h>
int main(void) {
    int n = 5, i, j, k;
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n - i; j++) {
            printf(" ");
        }
        for (k = 1; k <= i; k++) {
            printf("%d ", k);
        }
        printf("\n");
    }
}
```

```

        1
      1 2
    1 2 3
  1 2 3 4
1 2 3 4 5
```

```
#include <stdio.h>
int main() {
    int rows = 5;
    for (int i=0;i<rows;i++) {
        for (int j = 0; j<2 *
            (rows - i) -2; j++) {
            printf(" ");
        }
        for (int k=0;k<=i;k++) {
            printf("%d ", k+1);
        }
        printf("\n");
    }
    return 0;
}
```



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Pattern	Logic
<pre>Enter No Of Rows:5 1 2 2 3 3 3 4 4 4 4 5 5 5 5 5</pre>	<pre>#include <stdio.h> void main() { int rows, i, j; printf("Enter No Of Rows:"); scanf("%d", &rows); for(i=1; i<=rows; i++) { for(j=1; j<=i; j++) { printf("%d ", i); } printf("\n"); } }</pre>
<pre>Enter No Of Rows:5 1 2 3 4 5 2 3 4 5 3 4 5 4 5 5</pre>	<pre>#include <stdio.h> void main() { int rows, i, j; printf("Enter No Of Rows:"); scanf("%d", &rows); for(i=1; i<=rows; i++) { for(j=i; j<=rows; j++) { printf("%d ", j); } printf("\n"); } }</pre>



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Pattern

```
1 2 3 4 5 6 7 8 9
 1 2 3 4 5 6 7
   1 2 3 4 5
    1 2 3
     1
```

Logic

```
#include <stdio.h>
int main(){
    int rows = 5;
    for (int i=0; i<rows; i++) {
        for (int j=0; j<2*i; j++){
            printf(" ");
        }
        for (int k=0; k<2 *
(rows-i) -1; k++) {
            printf("%d ", k+1);
        }
        printf("\n");
    }
}
```

```
1 2 3 4 5
1       5
1       5
1       5
1 2 3 4 5
```

```
#include <stdio.h>
int main() {
    int rows = 5;
    for (int i=0; i<rows; i++) {
        for (int j=0; j<rows; j++) {
            if (i>0 && i<rows-1 && j>0
&& j<rows-1) {
                printf(" ");
            }
            else {
                printf("%d ", j + 1);
            }
        }
        printf("\n");
    }
    return 0;
}
```



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Pattern	Logic
<pre>Enter NO of Rows:5 1 1 2 1 2 3 1 2 3 4 1 2 3 4 5</pre>	<pre>#include <stdio.h> void main() { int rows, i, j; printf("Enter NO of Rows:"); scanf("%d", &rows); for(i=1; i<=rows; i++) { for(j=1; j<=i; j++) { printf("%d ", j); } printf("\n"); } }</pre>
<pre>Enter No Of Rows:6 6 6 5 6 5 4 6 5 4 3 6 5 4 3 2 6 5 4 3 2 1</pre>	<pre>#include <stdio.h> void main(){ int rows, i, j; printf("Enter No Of Rows:"); scanf("%d", &rows); for(i=rows; i>=1; i--) { for(j=rows; j>=i; j--) { printf("%d ", j); } printf("\n"); } }</pre>



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Pattern	Logic
<pre> 1 1 2 3 1 2 3 4 5 1 2 3 4 5 6 7 1 2 3 4 5 6 7 8 9 1 2 3 4 5 6 7 1 2 3 4 5 1 2 3 1</pre>	<pre>#include <stdio.h> int main() { int n = 5; for (int i=0; i<2*n-1;i++) { // the row number int comp; if (i < n) { comp = 2*(n-i)-1; } else { comp = 2*(i-n+1) + 1; } for (int j=0; j<comp; j++) { printf(" "); } for (int k=0; k<2*n-comp;k++){ printf("%d ", k + 1); } printf("\n"); } return 0; }</pre>

1. Simple Star Pattern:

```
#include <stdio.h>
int main() {
    int i, j;
    for(i=1; i<=5; i++) {
        for(j=1; j<=1; j++) {
            printf("*");
        }
        return 0;
    }
```

```
  *
 **
***
****
*****
```

Output:

```
**
***
****
*****
*****
```

2. Inverted Star Pattern:

```
#include <stdio.h>
int main() {
    int i, j;
    for(i=5; i<=1; i--) {
        for(j=1; j<=1; j++) {
            printf("*");
        }
        return 0;
    }
```

```
*****
****
***
**
*
```

Output:

```
****
***
**
*
*
```

3. Pyramid Pattern:

```
#include <stdio.h>
int main() {
    int i, j, k;
    for(i=1; i<=5; i++) {
        for(j=1; j<=5-i; j++) {
            printf("*");
        }
        for(k=1; k<=2*i-1; k++) {
            printf("*");
        }
        return 0;
    }
```

```
  *
 **
***
****
*****
```

Output:

```
*
***
****
*****
*****
*****
```

4. Diamond Pattern:

```
#include <stdio.h>
int main() {
    int i, j, k;
    for(i=1; i<=4; i++) {
        for(j=1; i<=4-i; j++) {
            printf("*");
        }
        for(k=1; k<=2*i-1; k++) {
```

```
  *
 ***
****
*****
*****
```

```
  *
 ***
****
*****
*****
```


1. Hollow Square Pattern:

```
#include <stdio.h>
int main() {
    int i, j, n;
    printf("Enter number : ' // // for loop for rows
    scanf("%d", &n);
    for(i=1; i<=n; i++) { // for loop for rows
        for(j=1; j<=n; j++) { // for loop for columns
            if(i==1 || i==n || j==1 || j==n)
                printf("* ");
            else
                printf(" ");
        }
    }
    return 0;
```

Output :

*	*	*	*	*	*
*	*	*			*
*	*	*			*
*					*
*	*	*	*	*	*

2. Floyd's Triangle:

```
#include <stdio.h>
int main() {
    int i, j, n, num = 1;
    printf("Enter the number of ' // for loop for rows
    scanf("%d", &n);
    for(i=1; i<=n; i++) { // for loop for rows
        printf("%d ", num);
    }
    return 0;
```

Output :

1
2 3
4 5 6
7 8 9 10

3. Pascal's Triangle:

```
#include <stdio.h>
int main() {
    int i, j, n, num;
    printf("Enter the number of rows : ' // for loop for rows
    scanf("%d", &n);
    for(i=0; i<=n; i++) { // for loop for rows
        num = 1;
        for(j=0; j<=i; j++) {
            num = num * (i-j) / (j+1);
        }
    }
    return 0;
}
```

Output :

1
2 3
4 5 6
7 8 9 10

4. Hollow Pyramid Pattern:

1. Hollow Square Pattern:

```
#include <stdio.h>
int main() {
    int i, j, n;
    n = 5;
    for(i=1; j<=5; i++){ // print stars at borders
        if(i==1 || i==n || j==1 || j==n) {
            printf("%d", j);
        }
    } return 0;
}
```

Output :

*	*	*	*	*	*	*	*
*							*
*							*
*							*
*							*
*	*	*	*	*	*	*	*

2. Floyd's Triangle Pattern:

```
#include <stdio.h>
int main() {
    int i, j, n, nm=1;
    for(i=1; i<=4; i++) { // for loop for rows
        printf("%c", -+);
    }
}
```

Output :

1
2 3
4 5 6
7 8 9 10
11 12 13 14 15

3. Pascal's Triangle Pattern:

```
#include <stdio.h>
int main() {
    int i, j, n=5; num;
    for(i=1; i<=4; j++) { // calculate
        printf("*"); // binomical coefficient
        num = num * (i-j+1) / j;
    }
} return 0;
```

Output :

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

3. Pascal's Triangle Pattern:

```
#include <stdio.h>
int main() {
    int i, j, n=5; num;
```

Output :

	1	
1	1	

1. Number Pattern (Increasing Triangle)

```
#include <stdio.h>
int main() {
    int i, j;
    for(i=1; i<=5; i++) { // for loop for rows
        for(j=1; j<=i; j++) { // for loop for columns
            printf("%d", j); // print numbers
        }
    }
    return 0;
}
```

Output:

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

2. Alphabet Pattern

```
#include <stdio.h>
int main() {
    int i, j;
    char ch = 'A'; // for loop for rows
    for(j=1; i<=4; j++) { // for loop for columns
        printf("%c", ch); // print numbers
    }
    return 0;
}
```

Output:

A
A B
A B C
A B C D

3. Right Angle Star Pattern (Reverse)

```
#include <stdio.h>
int main() {
    int i, j;
    for(i=5; i<=1; i--) { // for loop for rows
        for(j=1; j<=i; j++) { // for loop for columns
            printf("%");
        }
    }
    return 0;
}
```

Output:

**
*
*

3. Right Angle Star Pattern (Reverse)

```
#include <stdio.h>
int main() {
    int i, j;
    for(i=5; i<=1; i--) { // for loop for rows
        for(i=1; i<=i; i++) { // for loop for columns
```

Output:

++